

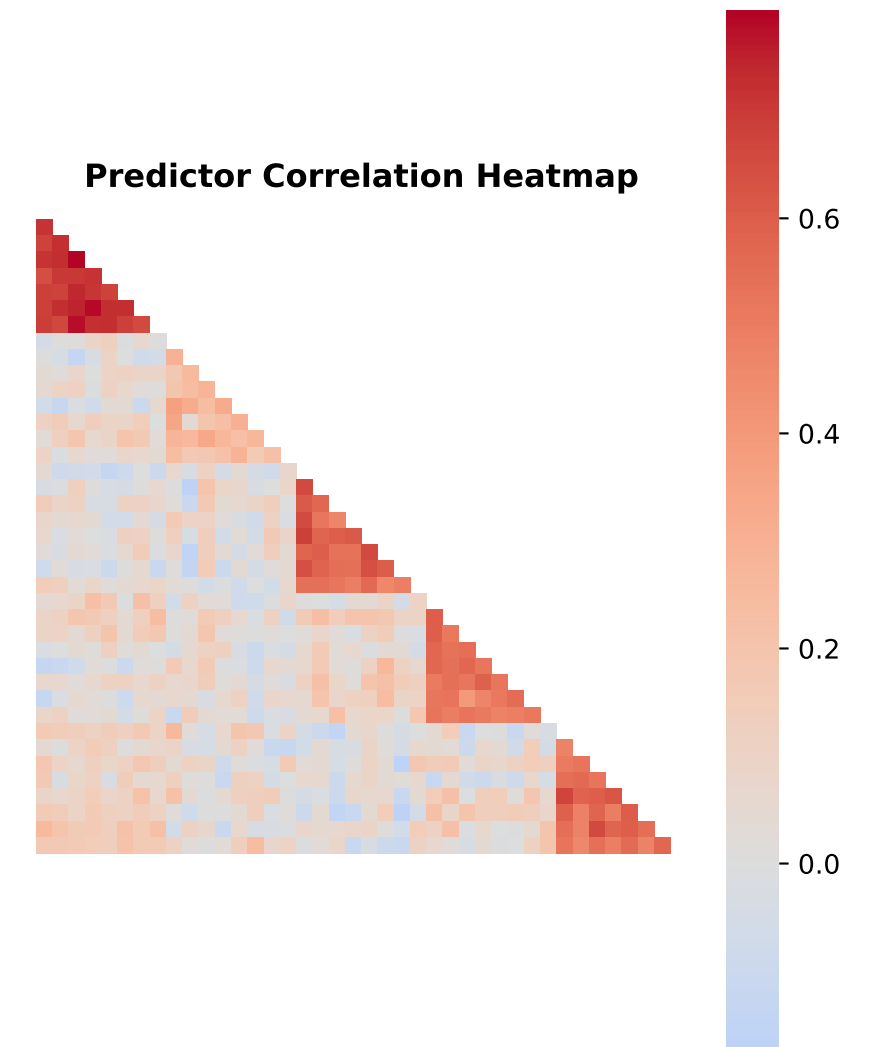
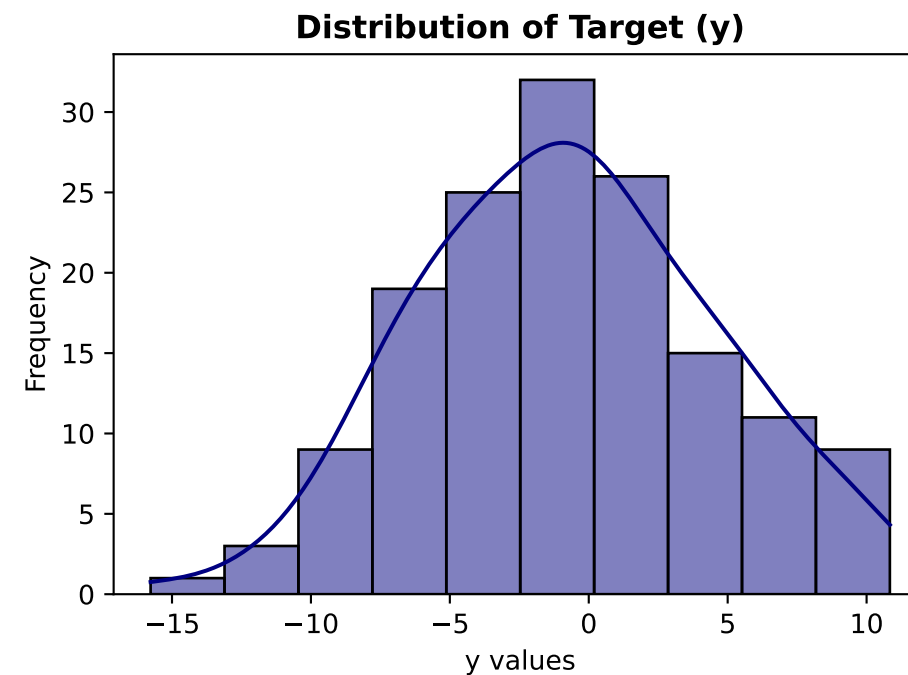
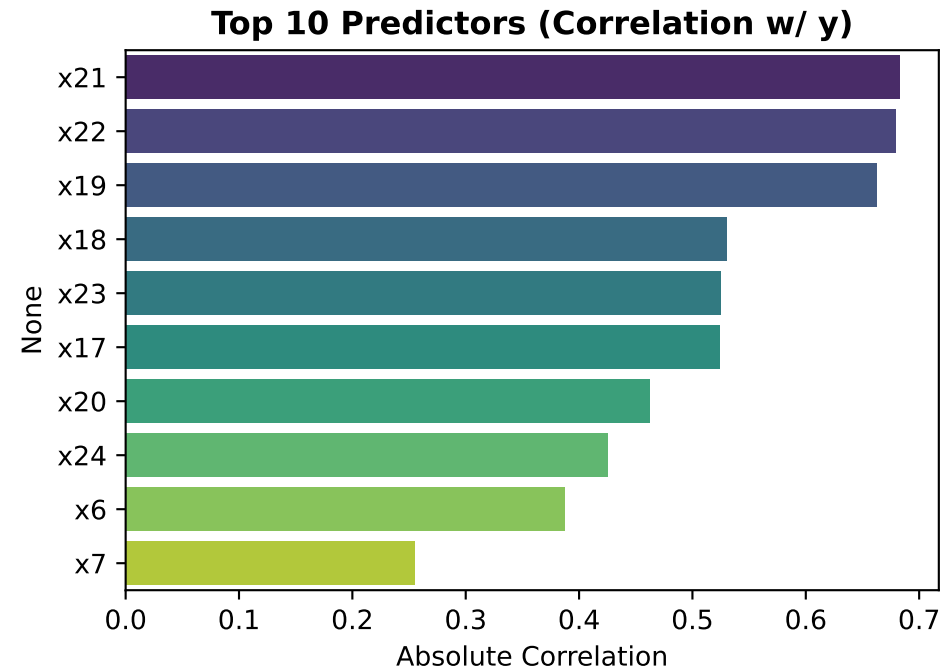
Data

Summary Statistics

Observations (n) : 150
Predictors (p) : 40

Target (y) Mean : -0.77
Target (y) Std : 5.27

Key Observation:
High-dimensional space with
heavy block-correlation requires
regularized model fitting.



Methodology & Results

Methodology

- Standardized all predictors to ensure fair penalization.
- Fit models using 5-Fold Cross-Validation to isolate optimal penalties.
- Evaluated both L2 (Ridge) and L1 (Lasso) regularization techniques.

Core Results

- Ridge Optimal Lambda: 0.0371
- Ridge variables retained: 40 (100%)
- Ridge mean test prediction: -1.19
- Ridge test prediction variance: 19.01

- Lasso Optimal Alpha: 0.1018
- Lasso variables retained: 18 (45.0%)
- Lasso mean test prediction: -1.13
- Lasso test prediction variance: 17.04

Final Ridge Model (Top 15)

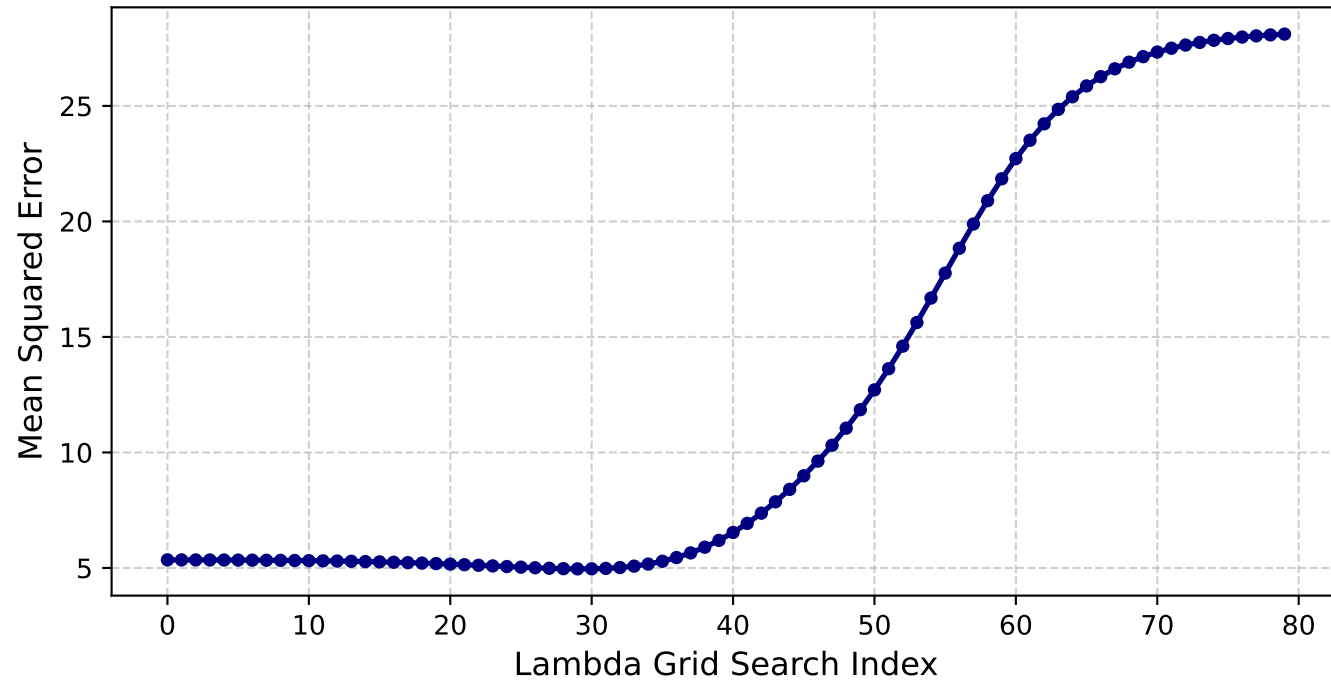
y_pred = intercept
+ (-2.2759 * x6)
+ (-2.0380 * x22)
+ (-1.6737 * x19)
+ (1.4699 * x2)
+ (1.2458 * x14)
+ (1.0588 * x29)
+ (-0.8335 * x21)
+ (-0.5621 * x39)
+ (-0.5422 * x11)
+ (-0.5157 * x18)
+ (0.4068 * x35)
+ (0.4043 * x17)
+ (0.3930 * x3)
+ (-0.3524 * x7)
+ (-0.3150 * x33)
... (+ 25 additional variables)

Final Lasso Model (Complete)

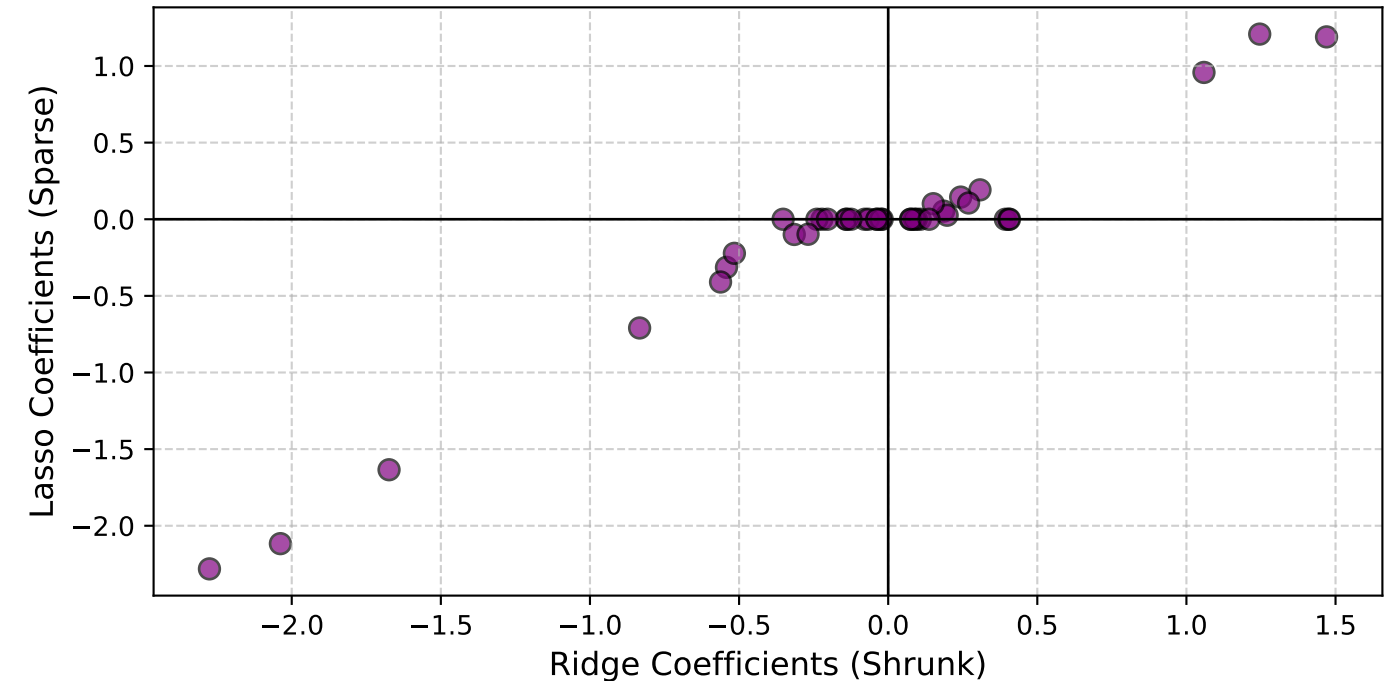
y_pred = intercept
+ (1.1899 * x2)
+ (-2.2814 * x6)
+ (0.0529 * x9)
+ (0.1445 * x10)
+ (-0.3139 * x11)
+ (0.0262 * x12)
+ (1.2081 * x14)
+ (0.1917 * x15)
+ (-0.2224 * x18)
+ (-1.6345 * x19)
+ (-0.7095 * x21)
+ (-2.1167 * x22)
+ (0.1061 * x27)
+ (0.9580 * x29)
+ (0.1014 * x30)
+ (-0.0993 * x33)
+ (-0.0980 * x34)
+ (-0.4095 * x39)

Visualization

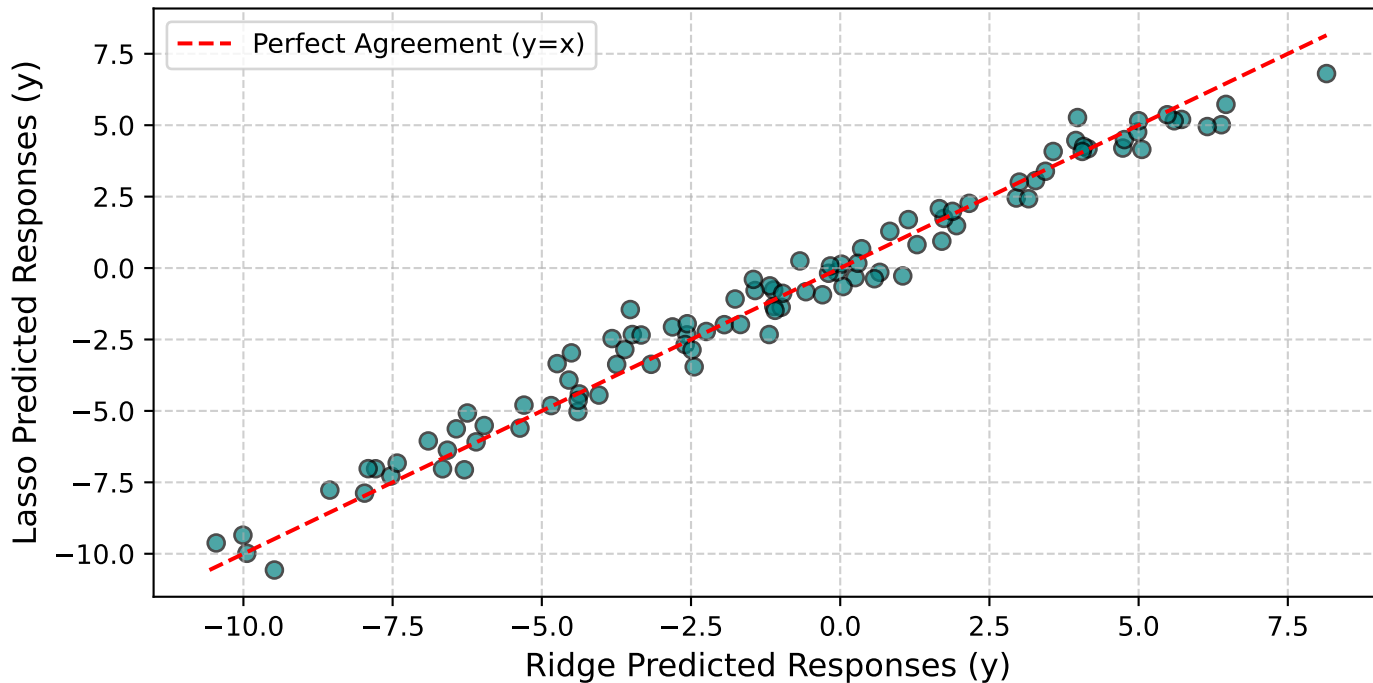
Methodology: Ridge Optimization (CV MSE Curve)



Results: Ridge vs. Lasso Coefficient Comparison



Results: Test Set Prediction Variance



Lasso Flaw: Bootstrap Selection Instability

